# System and Devices (SYS 3)

## Tutorial Lecture 1 : Instruction Pipeline & Performance

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# Amdahl's Law

A new floating-point unit speeds up floating-point operations by two times. In an application one fifth of the instructions are floating-point operations. (a) What is the overall speedup? (Ignore the penalty to other instructions). (b) Assume that the speeding up of the floating-point unit mentioned above slowed down data cache accesses resulting in a 1.5X slowdown. Assume that the load instructions constitute 15% and store instructions constitute 9% of the total instruction, what is the effective overall speedup now?

(a) $S = 1/\{(1-f) + (f/N)\} = 1/\{(1-0.2) + (0.2/2)\} = 1.11$ times

(b) $S = 1/\{(1-f1-f2) + (f1/N1) + (f2/N2)\} = 1/\{(1-0.2-0.24) + (0.2/2) + (0.24/0.67)\} = 0.98$ times

# Amdahl's Law

Suppose a processor spends 75% of its time in memory operations (both I-cache and D-cache access together) when running a given program A and its manufacturers make a change that improves its performance on memory access by a factor of 5. The program A originally took 100s to execute. What is the speed up from the old system to the new system? What fraction of its execution time does the new system spend in memory operations?

(a) $S = 1/\{(1-f) + (f/N)\} = 1/\{(1-0.75) + (0.75/5)\} = 2.5$ times

$S = Ext_old/Ext_new \quad 2.5 = 100/Ext_new$

$Ext_new = 40s$

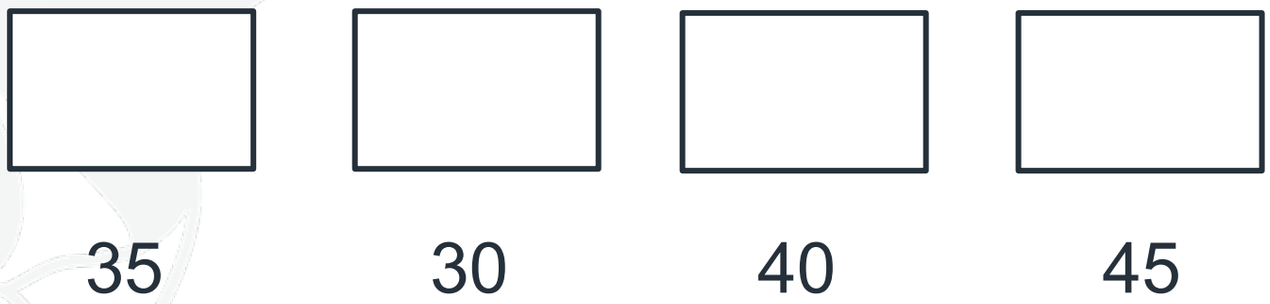
(b) Out of 100s in old system, 75% is memory → 75s

This 75s got speeded up by a factor of 5 → 75/5 = 15s

Hence, faction of memory in new is 15/40 = 37.5%

# Pipeline Design

The time delay of a 4-segment pipeline in different segments are t1=35ns, t2=30ns, t3=40ns, t4=45ns. The interface register delay time is t=5ns. How long would it take to complete 100 instructions in the pipeline? (Assume there is no dependency between the instructions).




$50 \times 100 = 5000$ns

$45 + 5 = 50$ ns

First instruction requires $50 \times 4 = 200$ns to complete

So, in total: $200 + 99 \times 50 = 5150$ns to complete

# Summary